

Nukabot: design of human-microbe-computer entanglement

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Figure 1: Physical appearance, interaction scene and system description of the Nukabot.

ABSTRACT

We present the latest development of Nukabot, a human-computer interaction mediation system that connects the fermented food microbiome with its human carer through voice communication. A *nukadoko* is a traditional Japanese fermentation technique that involves rice bran mixed with salt and water (*nuka*) put in a wooden, enamel, or plastic container (*doko*). The current version of Nukabot is a porcelain nukadoko equipped with chemical sensors to track the fermentation status, voice recognition, and vocal synthesis engines that enable humans to talk and ask questions about the ferment.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; **Empirical studies in interaction design**; **Empirical studies in HCI**; **HCI theory, concepts and models**.

KEYWORDS

entanglement; fermenting; microbes; sensing; longitudinal experiment; post-anthropocentric design

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1 INTRODUCTION

In recent years, HCI research has extended its reach to explore computer interaction with nonhuman actors, such as plants, animals, fungi, and microbes. Our work has investigated the possibilities of computer-mediated communication between humans and microbes residing in traditional fermented foods, drawing inspiration from posthumanistic discussions of more-than-human agencies in sociology, anthropology, philosophy, and STS (Science and Technology Studies) literature. We have designed Nukabot, a talkative fermented food vessel that enables humans to ask questions and be addressed by the microbial agency generated by a computational translation of chemical sensor values.

We note that Nukabot is a research product [Odom et al. 2016] and not a product proposal: its design aims to be *fit* to everyday life, *independent* so that it can be lived with for a long period without the researcher's intervention, and is driven by research *inquiries*, such as "how can technology help or inhibit embodied interaction between human and microbial agencies?" The authors are cautious not to iterate a prevalent technosolutionist design approach regarding both human and microbial agencies; instead, we explore the possibility of shaping technology that would nurture humans' bodily abilities to notice the world differently [Kuznetsov et al. 2011] by foregrounding microbial presence.

2 DESIGN & IMPLEMENTATION

Building upon our previous design iterations [Chen et al. 2019, 2021], the current version of the Nukabot consists of a physical body that integrates electronic parts connected to the Cloud to generate dynamic responses (Figure 1).

Masanori Jo, a ceramic artist residing in Kagoshima, Japan, crafted the body with porcelain in coordination with product designers Kiichi Moriya, Yuuto Mitani and Naoto Sekiya. The porcelain body consists of the body part in which fermented rice bran is placed and a lid to avoid the proliferation of aerobic bacteria. 3D printed

parts (gray colored) were attached to hide and protect the wirings of sensors and actuators. The blinking eyeball utilizes Nicodama, developed by media artist Ryota Kuwakubo and Perfektron LLC, which we rewired to control programmatically. The eyeball blinks every time it utters speech, which serves as a backchanneling element and social cue for generating a sense of naturalness to the conversation [Yoshimura et al. 2024].

The current electronic system consists of six types of sensors: four Atlas Scientific probes for sensing pH, ORP, electrical conductance, and soil temperature connected to a Raspberry Pi 4B, a Sseed Studio Grove Multichannel Gas Sensor (NH₃, CH₄, C₄H₁₀, NO₂, C₂H₅OH, CO, C₃H₈, and H₂ in parts per million values) and a temperature and humidity sensor connected to an Arduino Uno R4 unit, all connected to an Apple Mac mini (M1 silicon chip).

Every minute, each parameter value is sampled and recorded continuously in an Amazon Web Service RDS database (MySQL). A RESTful API built on a dedicated AWS EC2 server facilitated voice commands and responses. Speech recognition employs the open-source Efficient Word Net, with voice samples trained for the wake word "Ooi (Hey) Nukabot." Nukabot's distinctive voice is synthesized using the open-source library Open JTalk. Since November 2022, the Nukabot's speech ability has been powered by Rinna, a Japanese GPT transformer language model, to enable random conversation, such as small talk, beyond the preset Q&A functionality.

3 EXPERIMENTS

We have conducted several experiments with participants in addition to exhibitions in public venues. Our initial experiment, conducted in 2020, involved 6 participants who were asked to live with the Nukabot for 10 days. We have observed how participants' actions and attitudes of care [de La Bellacasa 2017] toward the ferment evolved through their daily interactions with the Nukabot [Chen et al. 2021].

From 2021 to 2023, we exhibited the Nukabot in Japanese public venues such as a science museum, design museum, and fermented food specialty store/restaurant while working on improving the system's stability. Each exhibition lasted 6 to 18 months, enabling the authors to observe the interaction patterns of over a hundred thousand visitors. Through these exhibition experiences, we noticed the important gap between the human and microbial actors and the Nukabot. While microbes constantly generate a dynamic fluctuation of fermentation and human perception evolves accordingly, the computational system remains stable. A new research question emerged: how can technological agencies such as software and hardware *ferment* through their interactions with humans and microbes?

4 DEMONSTRATION

To grapple with this inquiry, we have modeled a three-way entanglement of human, microbial, and computational actors as shown in Figure 2. Within this paradigm, the behavior of the Nukabot would change based on interferences from both humans and microbial activities. We have implemented two additional behaviors to the Nukabot to investigate this space of possibility. First, to reflect human influences, the Nukabot continuously learns new vocabulary

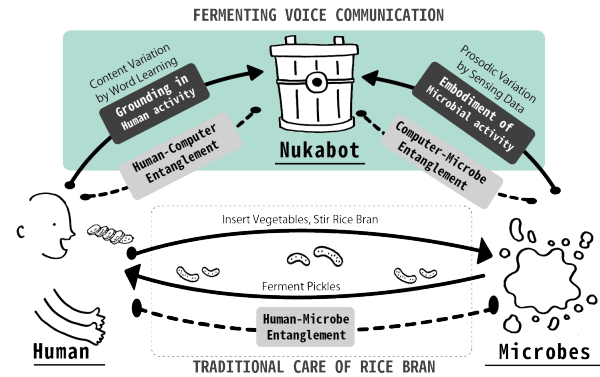


Figure 2: Entanglement model between human, microbial and computational agencies

from its conversation with the surrounding humans and incorporates these newly acquired words into GPT prompts. Secondly, the vocal prosody (pitch and speed) varies depending on the fermenting status of the rice bran. We will demonstrate these *fermenting* behaviors of the Nukabot with live, fermented rice bran and pickles.

5 CONCLUSION

We described the background and research-through-design processes of the Nukabot. In the future, we will further elaborate on the concept of *fermenting computational agency* by analyzing experiment results, leading to reconsidering the previous role of HCI designer-researcher when taking nonhuman and more-than-human agencies seriously.

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